

# Aerodynamic Design of a Mars UAV: Ingenuity Legacy

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## Mission Overview

On **19 April 2021**, NASA's Ingenuity helicopter became the first powered aircraft to fly on another planet, completing 72 flights before a rotor failure ended its mission in January 2024. The thin Martian atmosphere — just 1.6% of Earth's density — makes flight extremely challenging: rotors must spin at high RPM to generate lift, and the very-low Reynolds-number regime fundamentally changes airfoil aerodynamics.

This project develops the **first conceptual aerodynamic design** for a next-generation Mars UAV capable of carrying a **2 kg payload**, spanning hover power estimation, configuration selection, airfoil optimisation, and forward-flight wing analysis.

MARS GRAVITY

**3.73 m/s<sup>2</sup>**

38% of Earth

MARS AIR DENSITY

**0.020 kg/m<sup>3</sup>**

1.6% of Earth

DESIGN PAYLOAD

**2 kg**

vs Ingenuity: 0 kg

INGENUITY REFERENCE

**1.8 kg**

D = 1.2 m, 2800 rpm

## Task 1 — Ingenuity Power Baseline

Momentum theory applied to Ingenuity's coaxial counter-rotating rotor system. The lower rotor operates in the upper rotor's wake (interference factor  $f = 0.5$ ). A constant effective chord  $\bar{c} = 0.071$  m is derived from the published blade area.

### ① Induced velocity (upper rotor)

$$v_{i,u} = \sqrt{(T_u / 2\rho A)} = \sqrt{(3.357 / 2 \cdot 0.020 \cdot 1.131)} = \mathbf{8.61 \text{ m/s}}$$

### ② Profile power per rotor

$$P_p = \frac{1}{2} \rho \bar{c} N_b C_{d0} \Omega^2 R^4 = \mathbf{23.1 \text{ W}} \quad (C_{d0} = 0.02, \kappa = 1.15)$$

UPPER ROTOR

**56.4 W**

LOWER ROTOR

**43.7 W**

TOTAL SYSTEM

**≈ 100 W**

## Task 4 — Airfoil Selection for Low-Re Flight

### Reynolds number on Mars vs. Earth

$$Re_{\text{Mars}} / Re_{\text{Earth}} = (\rho/\mu)_{\text{Mars}} / (\rho/\mu)_{\text{Earth}} \approx \mathbf{0.028} \rightarrow \mathbf{36\times \text{ lower}}$$

$$Re \text{ at } r = 0.75R : (0.020)(133)(0.071) / 1.05 \times 10^{-5} \approx \mathbf{1.8 \times 10^4}$$

Viscous effects dominate; classical airfoils suffer early separation in this regime.

| Airfoil         | $C_{l,\text{max}}$ | Low-Re behaviour | Type            |
|-----------------|--------------------|------------------|-----------------|
| ROAMX-0201      | > 1.3              | Smooth, stable   | Mars-optimised  |
| 5% circular-arc | ~1.1               | Robust laminar   | Thin plate      |
| CLF5605         | ~0.9               | Higher drag      | Ingenuity blade |

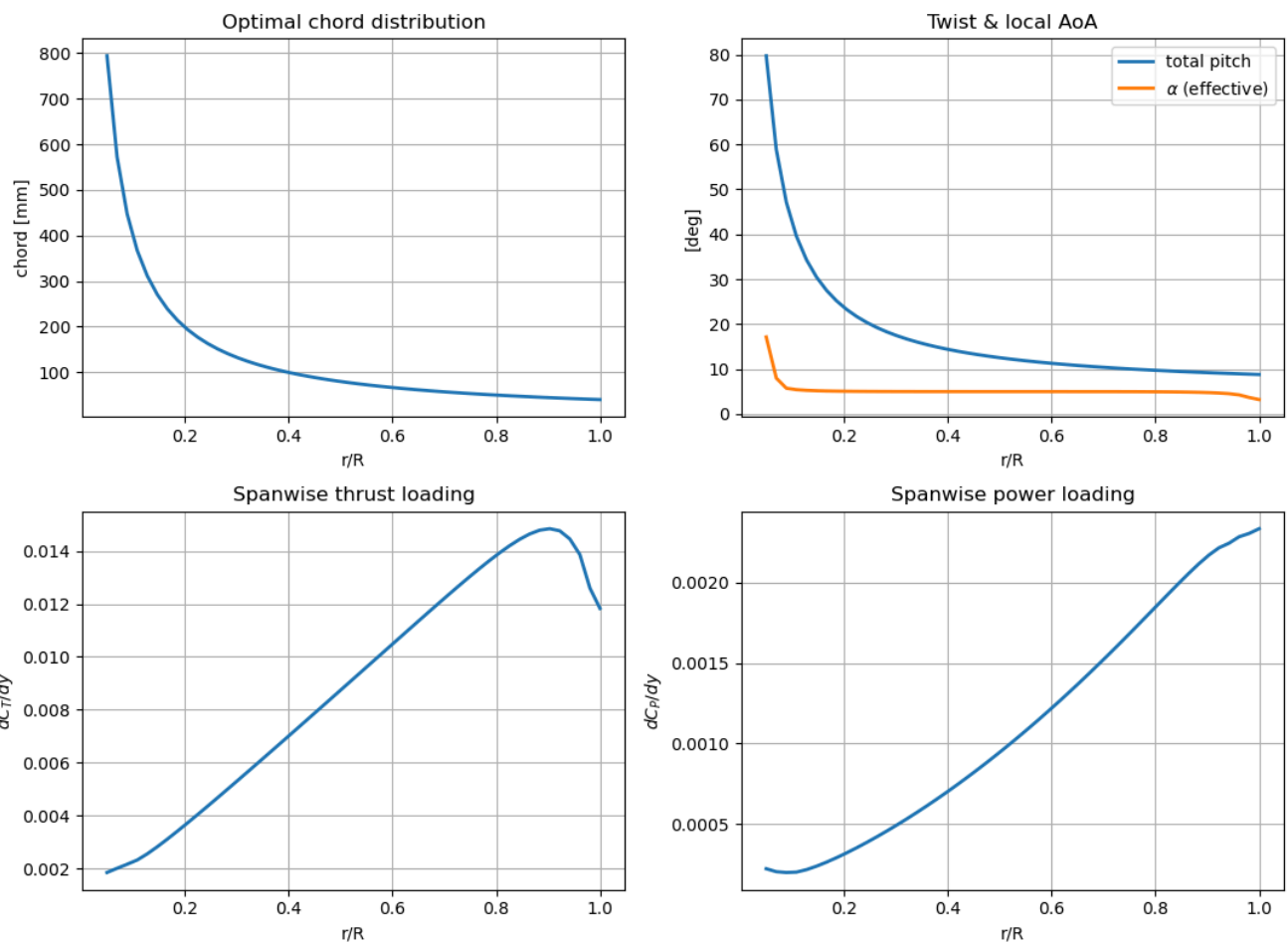


Fig. 1 —  $C_l$ - $\alpha$  and  $C_l$ - $C_d$  polars for candidate airfoils at  $Re \approx 1.8 \times 10^4$  (NASA literature data).

### Selected: ROAMX-0201

Highest  $C_{l,\text{max}} > 1.3$ , smooth stall behaviour, purpose-designed for Martian rotorcraft at very-low Reynolds numbers. No turbulator required for conceptual design.

## Task 2 — Conceptual Design & Configuration

Two configurations evaluated for hover with 2 kg payload. A coupled mass–power model scales Ingenuity's component weights (blades, motors, fuselage) with rotor radius and power. Optimal radius minimises total hover power.

| Parameter           | Twin-Rotor | Quadcopter ✓  |
|---------------------|------------|---------------|
| Number of rotors    | 2          | 4             |
| Optimal radius      | 0.69 m     | <b>0.59 m</b> |
| Blades per rotor    | 2          | 2             |
| Total hover power   | 197.3 W    | 105.4 W       |
| Flight time (hover) | 6.1 min    | 11.4 min      |

### Quadcopter selected — 47% lower power

Four rotors allow a smaller optimal radius per rotor, reducing induced and profile losses simultaneously. Power drops from 197 W to 105 W; flight time nearly doubles.

### Governing equations

$$P_t = \sqrt{(51.895 m_{\text{rot}}^2 / 2 N_{\text{rot}}^2 \rho \pi R^2)} \cdot \kappa$$

$$P_p = 178.7 R^4 \quad [\text{W, Mars, same } \Omega]$$

$$m_{\text{tot}} = 1.2(1500 + 29 R N_b N_{\text{rot}} + 2.5 P_{\text{tot}}) \quad [\text{g}]$$

ROTOR RADIUS

**0.59 m**

HOVER POWER

**105 W**

FLIGHT TIME

**11.4 min**

## Task 3 — Battery Optimisation

The 2 kg payload is replaced with extra battery cells ( $m_{\text{cell}} = 47$  g,  $E_{\text{cell}} = 10/6$  Wh). Two competing effects: more cells → more energy *and* more mass → higher power demand.

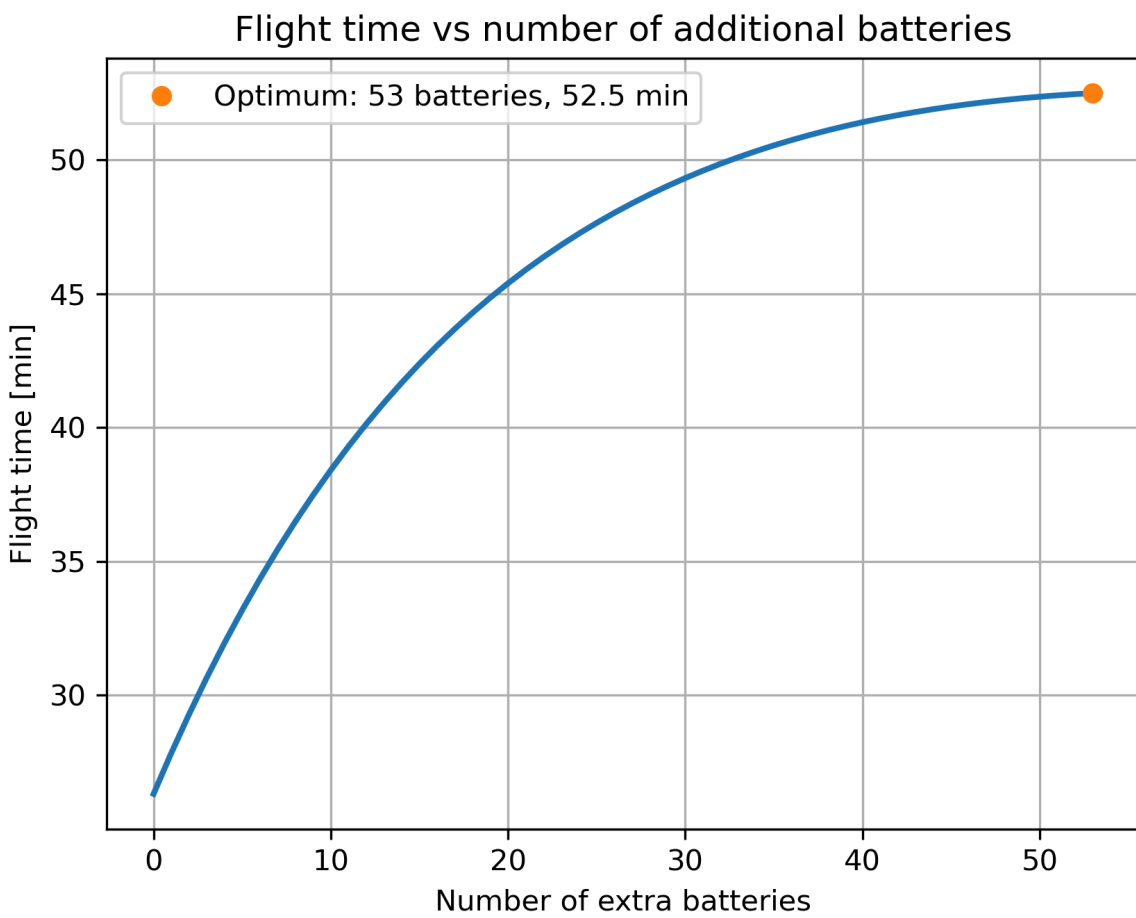


Fig. 2 — Flight time vs. number of extra battery cells. Energy gain consistently outweighs the mass penalty across the full 0–42 cell range.

OPTIMAL BATTERIES

**42**

MAX FLIGHT TIME

**51.7 min**

IMPROVEMENT

**+94%**

### Optimality condition

$$N_{\text{max}} = \lfloor 2 \text{ kg} / 0.047 \text{ kg} \rfloor = \mathbf{42 \text{ cells}}$$

$$\text{Battery mass: } 42 \times 47 \text{ g} = 1.97 \text{ kg} < 2 \text{ kg} \checkmark$$

$$t_{\text{flight}} = 60 \cdot E_{\text{tot}} / P_{\text{tot}} \approx \mathbf{51.7 \text{ min}}$$

Optimal = payload mass limit. Adding batteries always helps in this configuration.

| Configuration             | Energy         | Power          | Flight time     |
|---------------------------|----------------|----------------|-----------------|
| Original battery only     | 20.0 Wh        | 105.4 W        | 26.6 min        |
| <b>+ 42 battery cells</b> | <b>90.0 Wh</b> | <b>104.5 W</b> | <b>51.7 min</b> |

## Task 5 — Detailed Rotor Design (BEM)

BEM theory with Prandtl tip-loss corrections. Nelder–Mead optimizer (SciPy, 50 elements) minimises hover power over tip chord  $c_{\text{tip}}$ , design AoA  $\alpha_D$ , collective pitch  $\theta$ , and  $\Omega$ .

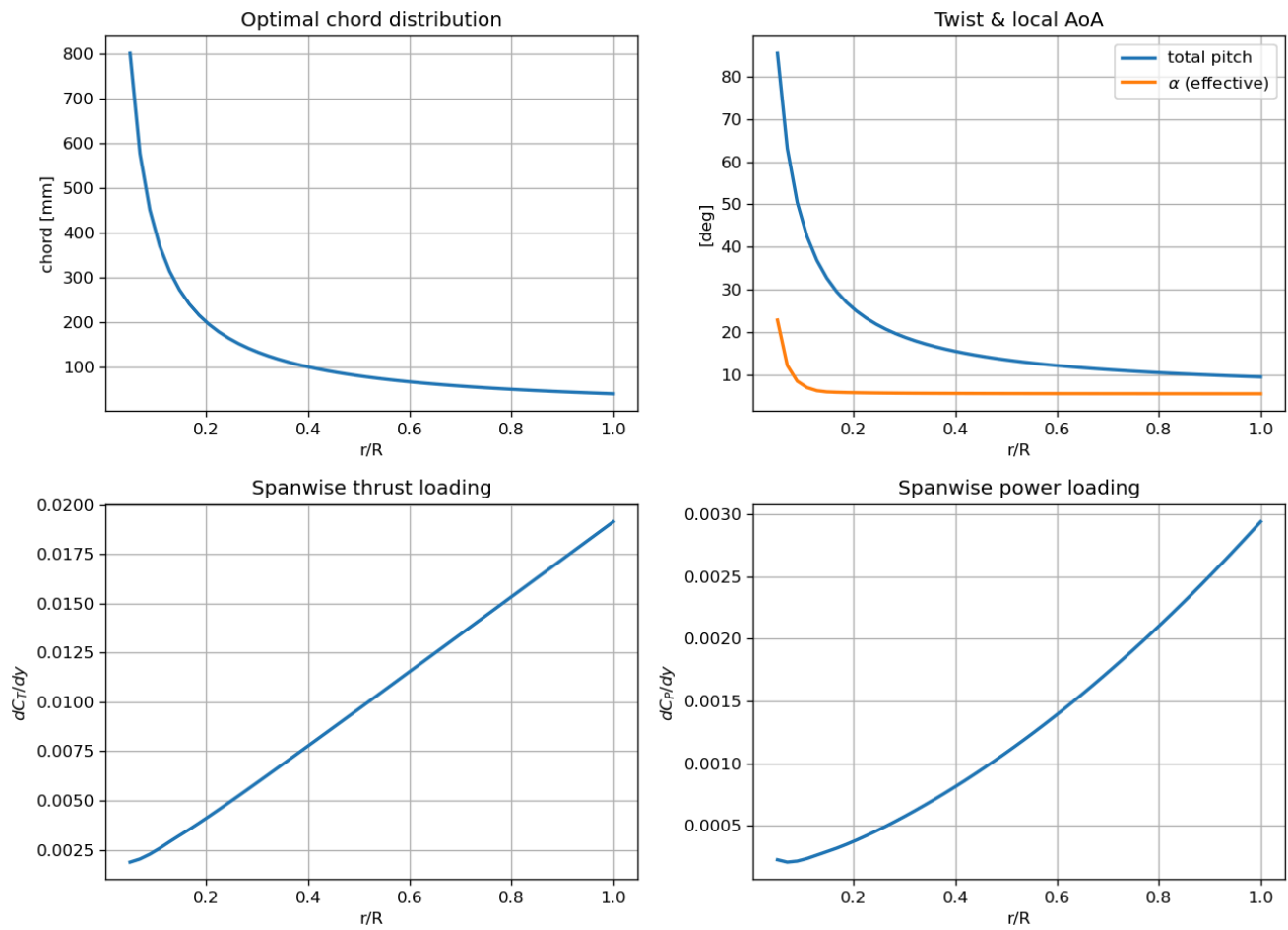


Fig. 3 — Optimised blade: chord, twist & local AoA, spanwise thrust  $dC_l/dy$ , and power loading  $dC_p/dy$ .

## Task 5 — Results

| Quantity                | Momentum Theory | BEM (Task 5)         |
|-------------------------|-----------------|----------------------|
| Method                  | Actuator disk   | Blade element + mom. |
| Power per rotor         | 26.4 W          | <b>107.7 W</b>       |
| <b>Total quad power</b> | <b>105.4 W</b>  | <b>430.8 W</b>       |
| Thrust per rotor        | —               | 4.27 N ✓             |

### BEM power is ~4× higher than momentum theory

Tip losses, hub losses, and blade profile drag absent from the actuator–disk model dominate real performance. Hyperbolic chord and high-twist root deliver required thrust with minimum power.

## Task 6 — Forward Flight & Wing Integration

### Mars dynamic pressure at $V = 10$ m/s

$$q = \frac{1}{2} \rho V^2 = \frac{1}{2} \cdot 0.020 \cdot 100 = \mathbf{1.0 \text{ Pa}}$$

≈ 60× lower than Earth at same speed — wing lift is intrinsically tiny on Mars.

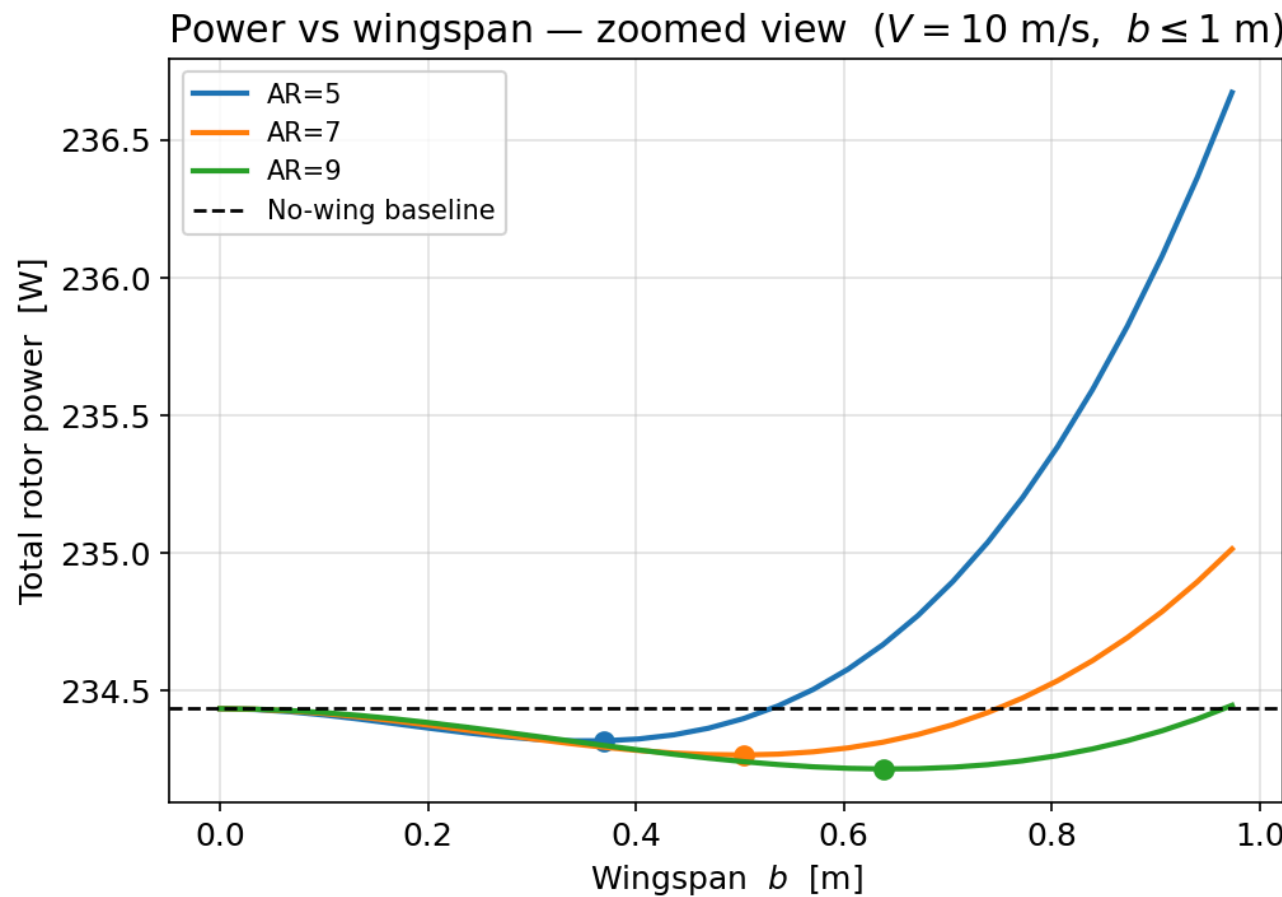


Fig. 4 — Total rotor power vs. wingspan (zoomed). Power barely dips below the no-wing baseline before wing mass dominates. Max reduction: **0.09%**.

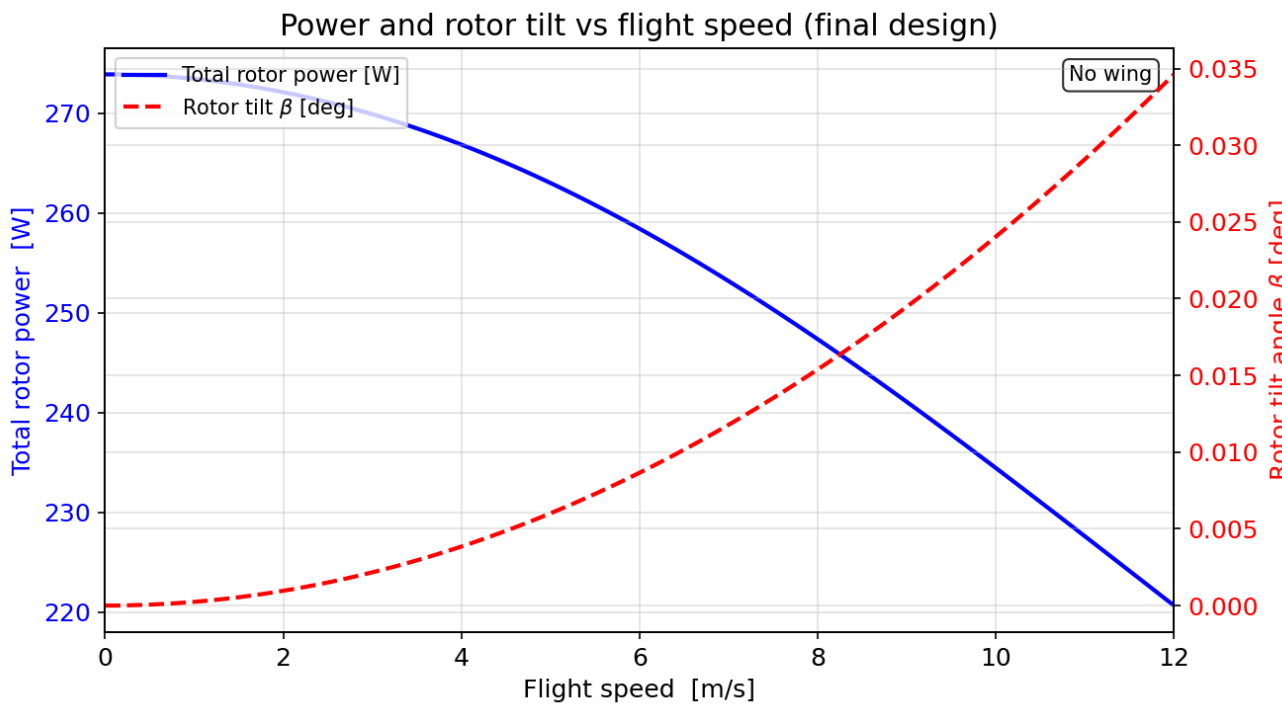


Fig. 5 — Total power and rotor tilt angle  $\beta$  vs. flight speed 0–12 m/s. Power dips below hover at low speeds (translational lift).

| Result                                       | Value                                                  |
|----------------------------------------------|--------------------------------------------------------|
| No-wing forward-flight power ( $V = 10$ m/s) | <b>234.4 W</b>                                         |
| LLT-optimal planform                         | Rect., $AR = 9$ , $-3^\circ$ washout, $e = 0.988$      |
| Best achievable power reduction              | <b>0.09%</b> (target: 10%)                             |
| Density for 10% reduction                    | $\rho_w \leq 7.0 \text{ kg/m}^3$ — <b>unachievable</b> |
| <b>Final design decision</b>                 | <b>Bare quadcopter — no wing added</b>                 |

## Conclusions

|               |                                                                                                                  |                                                                                   |
|---------------|------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------|
| <b>Task 1</b> | Ingenuity total rotor power ≈ 100 W via momentum theory — establishes the design baseline.                       |                                                                                   |
| <b>Task 2</b> | <b>Quadcopter selected</b> : 47% lower hover power (105 W vs. 197 W) and 87% longer flight time than twin-rotor. |                                                                                   |
| <b>Task 3</b> | <b>42 extra batteries</b> → 51.7 min (up from 26.6 min, +94%). Optimal at payload mass limit.                    |                                                                                   |
| <b>Task 4</b> | Re Mars ≈ $1.8 \times 10^4$ (36× below Earth). <b>ROAMX-0201</b> selected for best low-Re performance.           |                                                                                   |
| <b>Task 5</b> | BEM gives <b>107.7 W/rotor</b> — ~4× momentum estimate — due to tip and profile losses.                          |                                                                                   |
| <b>Task 6</b> | Wings impractical on Mars ( $q = 1.0$ Pa); max 0.09% reduction. Final design:                                    | <b>bare quadcopter</b> , $P = 430.8$ W, $t_{\text{fwd}} = 234$ W at $V = 10$ m/s. |

### Key takeaway

Mars's atmosphere ( $\rho = 1.6\%$  of Earth) forces large fast rotors at  $Re \approx 10^4$ , demands low-Re specific airfoils, and renders fixed wings impractical. BEM reveals that simplified momentum models underestimate real power by ~4×.

[1] Koning, W.J.F. et al. Mars Helicopter Rotor Design. NASA TM-2020-002224. NASA Ames Research Center, 2020.

[2] NASA. Mars Fact Sheet. nssdc.gsfc.nasa.gov/planetary/factsheet/marsfact.html (accessed 2026-04-14).

[3] NASA NTRS. Low Reynolds Number Airfoil Performance on Mars. ntrs.nasa.gov/citations/20240004230 (accessed 2026-04-14).

[4] Leishman, J.G. Principles of Helicopter Aerodynamics. Cambridge University Press, 2000.

[5] Balaram, B. et al. Mars Helicopter Technology Demonstrator. AIAA 2018-0023 (2018).